12.09.2026 September Questions

1 Real exam

The functions f and g are defined by

$$f(x) = x^2 + px + 42$$

and

$$g(x) = x^2 + 10x + 18,$$

where p is a constant. The solutions to $f(x) = 0$ are r and s, and the solutions to $g(x) = 0$ are $r + 2$ and $s + 2$. What is the value of p?

2 Real exam

Right circular cylinders A and B are similar. The height of cylinder A is **10 units**, and the diameter of its base is equal to its height. The ratio of the diameter of the base of cylinder A to the diameter of the base of cylinder B is

$$3 : 7.$$

What is the volume, in cubic units, of cylinder B?

A) 250π

B) $\dfrac{1{,}750\pi}{3}$

C) $\dfrac{12{,}250\pi}{9}$

D) $\dfrac{85{,}750\pi}{27}$

3 Real exam

The function f is defined by

$$f(x) = 3x^2 + 48x + 193.$$

The function g is defined by

$$g(x) = f(x + 7).$$

What is the minimum value of $g(x)$?

A) -15

B) -8

C) 1

D) 8

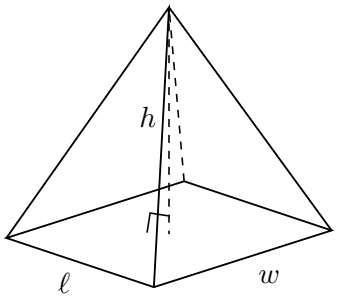
4 Real exam

The table summarizes the daily high temperatures, in degrees Fahrenheit, recorded in a town during the **31 days of December 2024**.

Daily high temperature, t (°F)	Frequency (days)
$10 < t \leq 15$	1
$15 < t \leq 20$	2
$20 < t \leq 25$	4
$25 < t \leq 30$	5
$30 < t \leq 35$	7
$35 < t \leq 40$	5
$40 < t \leq 45$	3
$45 < t \leq 50$	2
$50 < t \leq 55$	2

In January 2025, the daily high temperature in the same town was greater than 35°F on **18 days**. This number of days is p% greater than the number of days in December 2024 on which the daily high temperature was greater than 35°F. What is the value of p?

5 Real exam



Note: Figure not drawn to scale.

The figure shown is a right rectangular pyramid, where $\ell = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

6 Real exam

A sample of a certain isotope takes 29 years to decay to half its original mass. The function $s(t) = 128(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, that remains t years after a 128-gram sample starts to decay. Which statement is the best interpretation of $s(58) = 32$ in this context?

A) Approximately 32 grams of the sample remains 58 years after the sample starts to decay.

B) The mass of the sample has decreased by approximately 32 grams 58 years after the sample starts to decay.

C) The mass of the sample has decreased by approximately 58 grams 32 years after the sample starts to decay.

D) Approximately 58 grams of the sample remains 32 years after the sample starts to decay.

7 Real exam

The function f is defined by $f(x) = \dfrac{|x|}{a} - 14$, where $a < 0$. What is the product of $f(15a)$ and $f(8a)$?

8 Real exam

On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 21.5% of the square footage of the farmland, and there are buildings on exactly 14.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p?

9 Real exam

$$y < 24 - 8x$$
$$\frac{y}{8} > 9$$

Which inequality represents the x values for all solutions (x, y) that satisfy the given system of inequalities in the xy-plane?

A) $x > 3$

B) $x < 3$

C) $x > -6$

D) $x < -6$

10 Real exam

When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 8 resistors connected in series, where the resistance of each resistor is positive. This circuit has a total resistance of 130 ohms. The total resistance of 3 resistors in this circuit is 90 ohms. Which inequality best represents all possible values of the resistance x, in ohms, of one of the other 5 resistors in this circuit?

A) $0 < x < 8$

B) $0 < x < 40$

C) $40 < x < 90$

D) $40 < x < 130$

11 Real exam

$$\frac{x}{5}+\frac{y}{9}=\frac{47}{45}$$

An engineer connects resistors in series, where the resistors in the series have a total resistance of $\frac{47}{45}$ ohms. In this series, there are x resistors of type A, which each have a resistance of a ohms, and y resistors of type B, which each have a resistance of b ohms. The given equation represents this situation. According to this equation, what is the positive difference between the value of a and the value of b?

A) 47

B) 4

C) $\frac{47}{45}$

D) $\frac{4}{45}$

12 Real exam

A space probe uses a square-shaped solar sail to move through space. The side length of the solar sail is $8.89w$ meters, where w is the width, in meters, of the space probe. Which equation gives the area A, in square meters, of the solar sail?

A) $A=(w+8.89)(w+8.89)$

B) $A=(4)(8.89w)$

C) $A=(8.89w)(8.89w)$

D) $A=\left(\frac{8.89}{w}\right)\left(\frac{8.89}{w}\right)$

13 Real exam

A moving truck rental company charges its customers \$165 for the first hour of renting a truck and \$95 for each additional hour, plus the cost of gas used. Which equation represents this situation, where y is the total cost, in dollars, of renting a truck from this company for x hours, w is the cost, in dollars, of gas used, and $x \geq 1$?

A) $y=95(x-1)+w$

B) $y=165+95+w$

C) $y=165+95x+w$

D) $y=165+95(x-1)+w$

14 Real exam

The population density of Town A is 120 people per square mile, and the population density of Town B is 90 people per square mile. Town A has a population of 20,640 people, and Town B has a population of 3,870 people. The area, in square miles, of Town A is k times the area, in square miles, of Town B. What is the value of k?

15 Real exam

In **2010**, a skating team had a total of **51** skaters, each classified as either advanced or intermediate. From **2010** to **2020**, the number of advanced skaters on the team increased by approximately **59%**, and the number of intermediate skaters on the team increased by approximately **48%**. The total number of skaters on the team increased by approximately **53%**. Which equation best represents this situation, where a represents the number of advanced skaters on the team in 2010 and b represents the number of intermediate skaters on the team in 2010?

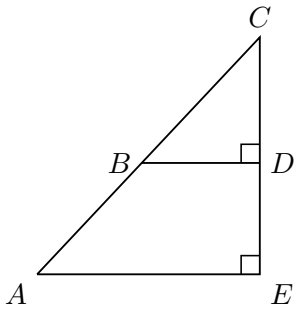
A) $1.48a + 1.59b = 51(1.53)$

B) $1.59a + 1.48b = 51(1.53)$

C) $1.53a + 0.59b = 51(1.48)$

D) $1.59a + 1.53b = 51(1.48)$

16 Real exam



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD. The measure of angle CBD is $57°$, and $AE = 25(BD)$. What is the measure of angle CAE?

A) $(25 \cdot 57)°$

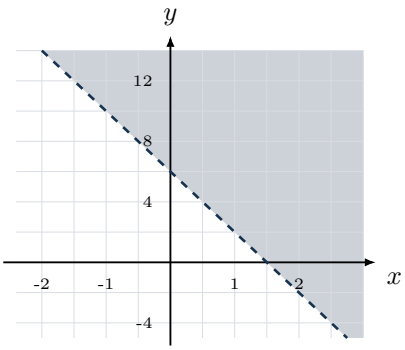
B) $(25 + 57)°$

C) $57°$

D) $25°$

17 Real exam

The shaded region shown represents the solutions to which inequality?



A) $y < 6x - 4$

B) $y > -4x + 6$

C) $y > 6x - 4$

D) $y < -4x + 6$

18 Real exam

In a certain state, the population of pheasants is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counted per mile of road decreased by 3.5% of the number of pheasants per mile of road the previous year. Based on this average, if there were 6.22 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

A) $0.965(6.22)^{10}$

B) $1.035(6.22)^{10}$

C) $6.22(0.035)^{10}$

D) $6.22(0.965)^{10}$

19 Real exam

When several resistors are connected in series in a circuit, the total resistance of the circuit is the sum of the resistances of these resistors. A certain circuit consists of 9 resistors connected in series, where the resistance of each resistor is positive. The total resistance of 4 resistors in this circuit is 60 ohms. Each of the other 5 resistors in this circuit has a resistance of at most 25 ohms. The total resistance of this circuit is x ohms. Which inequality best represents this situation?

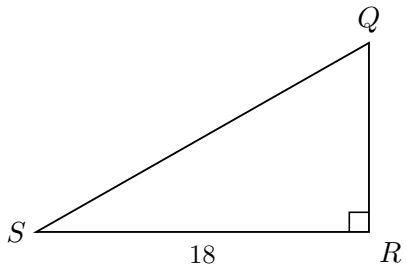
A) $0 < x \leq 185$

B) $60 < x \leq 125$

C) $60 < x \leq 185$

D) $x \geq 125$

20 Real exam



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

A) $18\cos Q$

B) $18\sin Q$

C) $\dfrac{18}{\cos Q}$

D) $\dfrac{18}{\sin Q}$

21 Real exam

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 2.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A) 3.78

B) 35.78

C) 487.89

D) 519.89

22 Real exam

In isosceles triangle ABC, sides AB and AC are congruent. Point D divides BC such that the length of $\overline{BD}$ is $\frac{5}{7}$ of the length of $\overline{BC}$. Point E lies on side AB and point F lies on side AC such that when segments DE and DF are drawn, angle BED is congruent to angle CFD. If the length of $\overline{BE}$ is 10, what is the length of $\overline{CF}$?

A) 4

B) 14

C) 50

D) 70

23 Real exam

In the xy-plane, a circle has center C with coordinates (h, k). Points A and B lie on the circle. Point A has coordinates $\left(h+1,\ k+\sqrt{102}\right)$, and $\angle ACB$ is a right angle. What is the length of $\overline{AB}$?

A) $\sqrt{206}$

B) $2\sqrt{102}$

C) $103\sqrt{2}$

D) $103\sqrt{3}$

24 Real exam

The water depth in a certain river is 78.7 feet (ft). During a 30-day period, the water depth in this river is predicted to increase by a minimum of 1 ft and a maximum of 3 ft. What is the predicted maximum water depth, in feet, during this time?

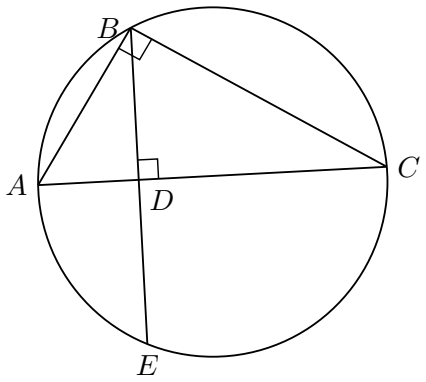
A) 75.7

B) 77.7

C) 79.7

D) 81.7

25 Real exam



Note: Figure not drawn to scale.

In the figure shown, points A, B, C, and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D, and $BD = \sqrt{346}$. The diameter of the circle is 175. If $\dfrac{CD}{AD} = r$, what is the value of r?

26 Real exam

The composition of an animal is defined as the muscles, bones, and fat of the animal. A scientist studied the composition of one young swamp buffalo and determined the buffalo had 133 kilograms of muscle, which made up approximately 65.9% of its composition. Of the remaining composition of this buffalo, approximately 46.2% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone?

27 Real exam

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 56 square centimeters (cm^2), and the surface area of right rectangular prism Y is 896 square centimeters (cm^2). The volume of right rectangular prism Y is 1,536 cubic centimeters (cm^3). What is the sum of the volumes, in cm^3, of right rectangular prism X and right rectangular prism Y?

28 Real exam

A clothing store buys shirts at a wholesale price of 5.00 dollars each and resells them each at a price that is 270% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 70% of the retail price. What is the discounted price of each remaining shirt, in dollars?

29 Real exam

The function g is defined by $g(x) = -\sqrt{x^2 + bx + c}$, where b and c are integers. In the xy-plane, the graph of $y = g(x)$ passes through the points $(2, 0)$ and $\left(0, -\sqrt{299}\right)$. If m is a value for which $g(m) = 0$, what is the greatest possible value of m?

A) -150

B) 2

C) 148

D) 150

30 Real exam

Each 3.0-ounce serving of cheddar cheese and each 1.2-ounce serving of tuna provides about 1 microgram of vitamin B12. If a total of 3.5 micrograms of vitamin B12 are consumed from eating x ounces of cheese and y ounces of tuna, which equation best represents this situation?

A) $0.33x + 0.33y = 3.5$

B) $3.0x + 1.2y = 3.5$

C) $1.2x + 3.0y = 3.5$

D) $0.33x + 0.83y = 3.5$